

BRAVO Lab

Brazilian Risk, Allocation, Volatility & Options Lab

Institutional Executive Report

PRIVATE APPLICATION DRAFT · v0.2.1

A reproducible Python research prototype for Brazilian market stress, derivatives-overlay logic, active-risk diagnostics, and portfolio-governance review.

CREATED BY

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Research, portfolio risk, derivatives overlays, and market transmission analysis

EXECUTIVE SUMMARY

BRAVO Lab is not a trading signal. It is a decision-support prototype.

BRAVO Lab converts Brazilian market stress into a structured portfolio-governance conversation. Its purpose is not to predict the market. Its purpose is to make exposure decisions easier to inspect: when to keep passive beta, when to harvest volatility income, when to protect downside risk, and when the evidence is too weak to justify action.

The project's value is discipline. It links data processing, stress indicators, derivatives-overlay logic, active-risk diagnostics, visual reporting, and validation limits into one reproducible workflow. That is more credible than a single impressive chart because it shows how a financial research idea can become decision material.

Core read

The strongest professional framing is simple: BRAVO Lab is applied Python for Brazilian market stress, derivatives overlays, and portfolio-governance review.

WHAT BRAVO LAB IS

A research workflow built around the portfolio question.

BRAVO Lab is a Python-based research prototype for Brazilian market risk. It connects B3 equity exposure, volatility regimes, derivatives overlays, stress-transmission indicators, and executive reporting. The system is not positioned as a finished investment engine. It is positioned as an inspectable research infrastructure.

The central question is practical: what should a portfolio do when Brazilian market stress changes state? A passive strategy keeps beta. A covered-call overlay sells part of the upside for income. A collar pays for protection. A stress-aware policy tries to connect those choices to regime evidence.

Why this matters

The project becomes valuable when it turns technical output into a reviewable decision, not when it claims certainty.

NU / NUBANK RELEVANCE

The asset is not only the PDF. The asset is the workflow behind it.

For a Nu/Nubank reviewer, the strongest signal is not that BRAVO Lab is complete. The strongest signal is that it demonstrates initiative, applied Python, Brazilian market intuition, derivatives reasoning, risk communication, and research honesty.

The project shows that complex financial ideas can be translated into structured evidence: data files, charts, reports, validation limits, and executive language. That combination matters because real financial work is rarely just modeling. It is also explanation, governance, and trust.

Application angle

BRAVO Lab should be presented as evidence that you can build financial intelligence material that a risk or investment team can challenge.

PORTFOLIO GOVERNANCE PROBLEM

The real problem is not volatility. It is what the portfolio is allowed to do when volatility changes.

Many reports stop at describing volatility. BRAVO Lab should go one step further. The more important issue is governance: when stress rises, should the portfolio remain passive, collect premium, buy protection, or wait for stronger confirmation?

That question is institutional because it forces trade-offs into the open. Income is not free. Protection has a cost. Switching can create false positives. Passive exposure can be elegant in calm periods and painful when stress persists. BRAVO's job is to make those trade-offs visible.

Decision principle

A useful model does not remove judgment. It organizes judgment so that it can be discussed, tested, and improved.

METHOD ARCHITECTURE

From raw market evidence to decision material.

The BRAVO workflow begins with market data and processed indicators. Those indicators feed stress diagnostics, strategy comparison, drawdown analysis, and BSTI policy logic. The final output is not just a table of results. It is a report that explains what the results mean for portfolio review.

This architecture matters because it creates a bridge between technical research and professional communication. A recruiter or portfolio reviewer does not need to read every line of code first. The report should reveal the logic, while the repository provides the evidence behind it.

Architecture read

The system is strongest when every page can be traced back to a file, chart, or assumption inside the project.

EVIDENCE DASHBOARD

The dashboard should summarize the current risk state without pretending to be the whole model.

This page is the fastest review surface in the report. It condenses the current stress read, regime label, policy output, and key portfolio diagnostics into a single visual checkpoint.

BRAVO Lab Executive Risk Dashboard

Brazilian risk, allocation, volatility, options, and stress-transmission intelligence

Latest BSTI

15.17

Composite stress score, 0 to 100

BSTI regime

Fragile

Current transmission state

Dominant channel

Brazil Drawdown Pressure

Main pressure source

Policy action

Covered Call

Current overlay rule

Best information ratio

Bsti Policy Overlay

Highest active-risk efficiency

Best drawdown profile

Collar

Least severe max drawdown

Source: BRAVO Lab processed outputs. Dashboard is generated automatically from reproducible CSV evidence files.

Portfolio read

The dashboard is useful only if it leads to a portfolio question: does the current evidence justify changing the Brazil equity overlay, or is the signal not strong enough to act?

STRATEGY COMPARISON

The point is not to find a magic strategy. The point is to make the trade-off visible.

BRAVO Lab compares strategy behavior as a governance problem. Passive beta, covered-call income, collar protection, and stress-aware switching each solve a different problem. A serious portfolio discussion begins when those trade-offs are made explicit.

LAYER	INSTITUTIONAL READ
Passive beta	Keeps full upside participation, but leaves the portfolio exposed when stress becomes persistent.
Covered call	Harvests volatility income, but can sell recovery too cheaply if the market rebounds quickly.
Collar	Improves downside discipline, but introduces protection cost and can limit participation.
BSTI policy	Creates a review rule that connects stress evidence to action, but must be validated against false positives and trading friction.

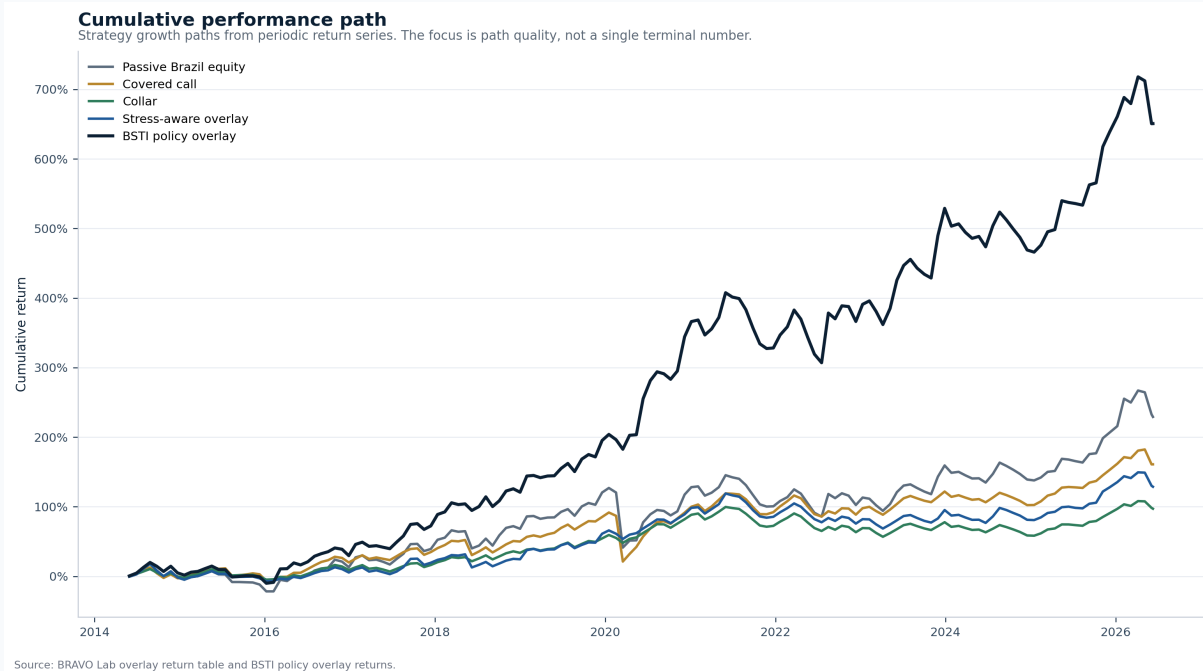
Governance read

The best strategy is not universal. The right overlay depends on regime persistence, option cost, liquidity, drawdown tolerance, and the mandate's tolerance for tracking error.

PERFORMANCE EVIDENCE

Performance is only useful when the trade-off is visible.

The cumulative path shows how the tested strategy layer behaves over time. The chart should not be read as a victory lap. It should be read as a prompt to ask what risks were taken to produce the path.



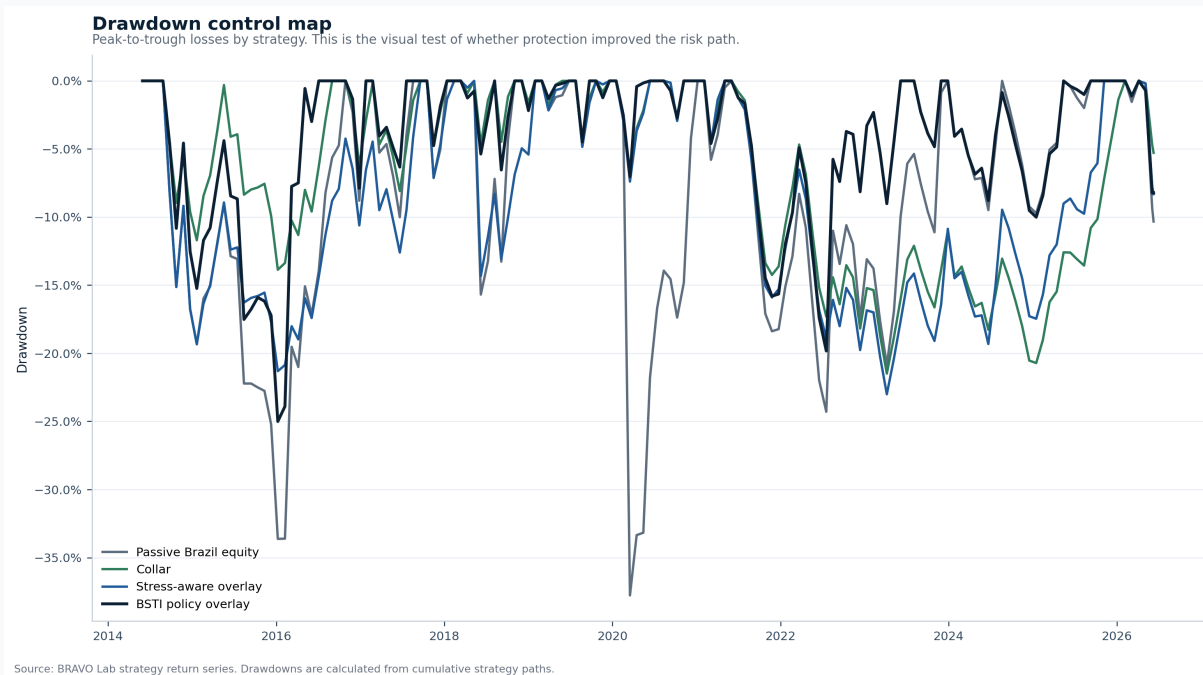
Portfolio read

A portfolio reviewer should ask whether the return profile is still attractive after considering drawdown, tracking error, option cost, turnover, and implementation friction.

RISK CONTROL

Drawdown is where the overlay earns or loses credibility.

Drawdown evidence matters because it shows what happens when the portfolio moves away from peak value. A strategy that looks good in cumulative performance can still fail if it does not control stress periods.



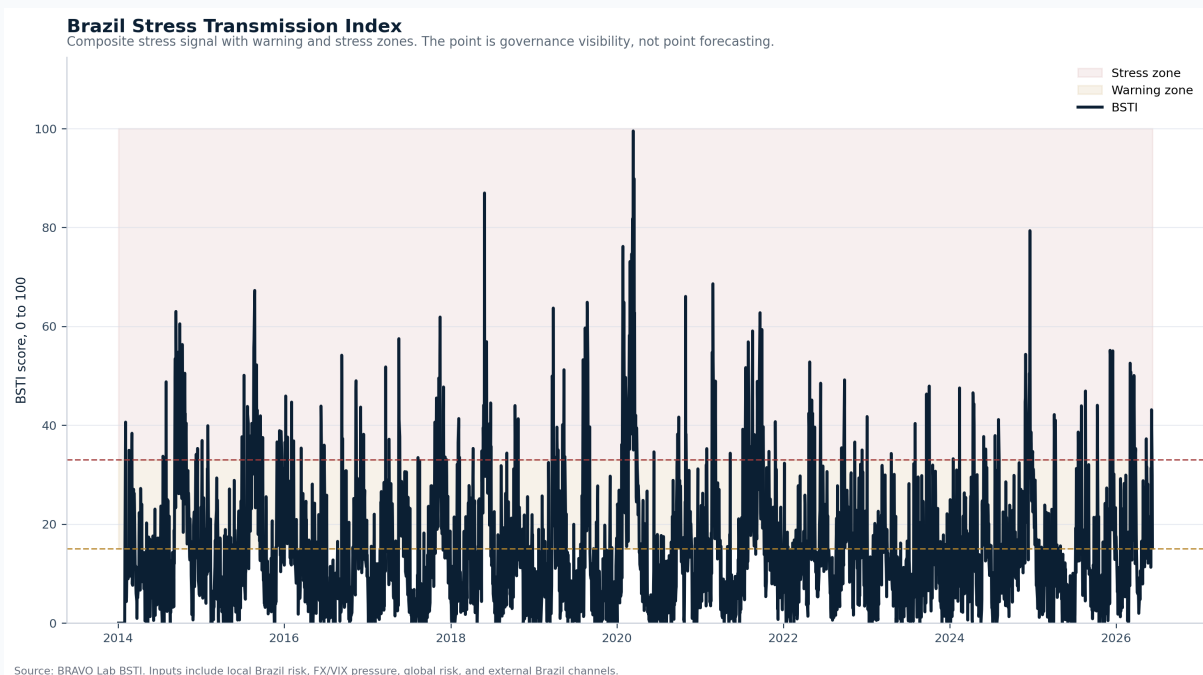
Portfolio read

The relevant question is whether the overlay reduces pain enough to justify its cost, complexity, and possible sacrifice of upside participation.

BSTI SIGNAL LOGIC

A stress index is useful only when it improves the review discipline.

The Brazil Stress Transmission Index should be treated as a review trigger, not a prophecy. Its value is in helping the reviewer separate ordinary noise from conditions that deserve closer attention.



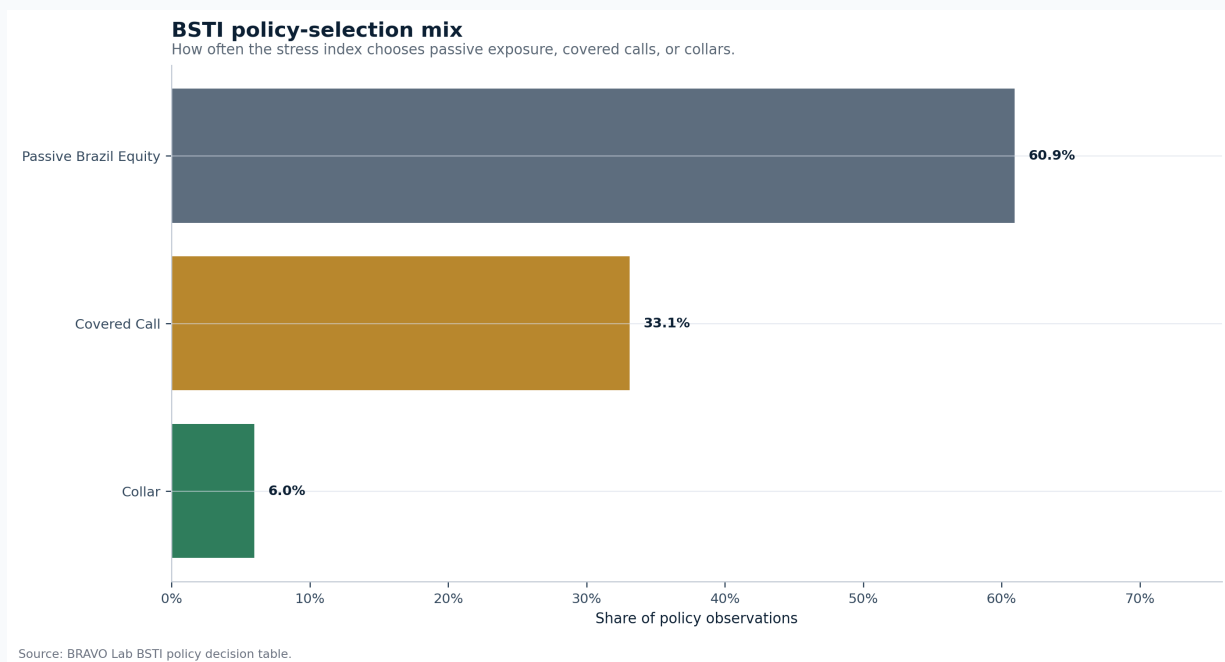
Portfolio read

The professional question is where the threshold should sit: when does the signal justify observation, hedge review, income overlay, or downside protection?

POLICY LOGIC

Policy selection is the bridge between signal and action.

The policy-selection mix reveals whether the system is mostly passive, defensive, income-oriented, or too interventionist. That matters because excessive switching can be as damaging as ignoring stress.



Portfolio read

The key issue is discipline: does the rule change behavior only when the evidence is strong enough, or does it create unnecessary trading noise?

VALIDATION LIMITS

The most credible page is the one that admits what still needs to be tested.

BRAVO Lab should not hide its limits. The current prototype is useful as application evidence and research infrastructure, but the derivatives layer still needs observed B3 option-chain validation. Synthetic assumptions are acceptable for a prototype, not for a final implementation claim.

The next credibility layer should include real strikes, maturities, bid-ask spreads, liquidity filters, transaction costs, taxes, roll rules, and execution assumptions. Without that layer, the project should remain framed as decision-support research, not as an invest remain framed as decision-support research, not as an investable strategy.

Credibility read

Professional honesty strengthens the project. It shows that the model is designed to be challenged, not sold as magic.

NEXT DEVELOPMENT ROADMAP

The next build should increase trust, not complexity.

LAYER	INSTITUTIONAL READ
1. Option chain	Replace synthetic option assumptions with observed B3 strikes, maturities, spreads, and liquidity filters.
2. Frictions	Add transaction costs, taxes, roll frequency, execution assumptions, and portfolio turnover diagnostics.
3. Volatility	Introduce MTV-GARCH, DCC, or related covariance logic to strengthen risk estimation.
4. Stress blocks	Use CCA or Ridge CCA to identify which market blocks transmit stress most clearly.
5. Multiscale layer	Use Airy or Generalized Morse Wavelets to separate tactical noise from persistent stress.
6. Application note	Create a short Nu/Nubank-facing note that explains the project in recruiter language.

Build principle

Every new layer should answer one professional question: does it make the portfolio decision more transparent, more testable, or more useful for risk review?

DEEP SWOT

The project becomes stronger when the limits are part of the argument.

Strengths

End-to-end Python workflow, Brazil-specific market framing, derivatives-overlay intuition, reproducible evidence, and a decision-oriented reporting layer.

Weaknesses

The current option layer still needs real B3 option-chain validation. Transaction costs, liquidity filters, and tax assumptions must be made explicit.

Opportunities

The next version can integrate observed option chains, MTV-GARCH or DCC volatility dependence, CCA stress compression, and wavelet-based multiscale analysis.

Threats

The project loses credibility if it is framed as prediction, investment advice, or a finished production engine. Its correct role is decision-support evidence.

EVIDENCE MAP

A serious prototype should be easy to inspect.

BRAVO Lab is strongest when the narrative points back to reproducible evidence. The PDF should not replace the project files. It should guide the reviewer toward the code, reports, figures, and processed outputs that support the argument.

LAYER	INSTITUTIONAL READ
Executive memo	reports/front_office_memo.md
Technical report	reports/baseline_report.md
Figures	reports/figures/
Processed outputs	data/processed/
PDF generator	scripts/generate_institutional_pdf.py
Application note	reports/nu_asset_application_note.md

Final positioning

BRAVO Lab should be presented as a portfolio-governance research prototype: serious enough to inspect, honest enough to challenge, and clear enough to discuss in an interview.

BRAVO Lab

From Brazilian market stress to portfolio-governance review.

Research prototype · Private application evidence · Not investment advice

Created by Rodolfo Pereira